

UNIFIED MENTOR

INTERNSHIP PROGRAM 2026

PROJECT SUBMISSION REPORT

DOMAIN: MACHINE LEARNING

PROJECT COMPLETED

European Bank Analytics: Customer Churn Pattern & Segmentation

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PROJECT REPORT

Customer Segmentation & Churn Pattern Analytics

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[View Code: GitHub Repository](#)

1. ABSTRACT

This project leverages historical European banking data (10,000 records) to identify key factors contributing to customer churn. By analyzing demographics, geography, and financial profiles, it isolates high-risk segments. The solution utilizes Python for data manipulation and Streamlit for an interactive, enterprise-grade analytical dashboard.

2. PROBLEM STATEMENT

Customer churn represents systemic capital flight and reduced lifetime value. The challenge lies in moving beyond generic churn rates to uncover granular insights: which demographics are leaving, how churn differs by country, and identifying wealth flight among premium account holders.

3. PROJECT OBJECTIVES

- 1. Risk Identification:** Measure overall churn rates and isolate high-risk customer segments.
- 2. Geographic Analysis:** Compare churn behaviors across France, Spain, and Germany.
- 3. Wealth Flight Tracking:** Quantify the financial profile of churned vs. retained customers.
- 4. Live Dashboarding:** Build a dynamic Streamlit interface for live KPI monitoring and drill-downs.

4. TOOLS & METHODOLOGY

- 1. Stack:** Python, Streamlit, Pandas, Plotly Express.
- 2. Process:** Data validation, segmentation engineering (Age bands, Balance tiers), and dynamic UI/UX design via custom CSS injection.

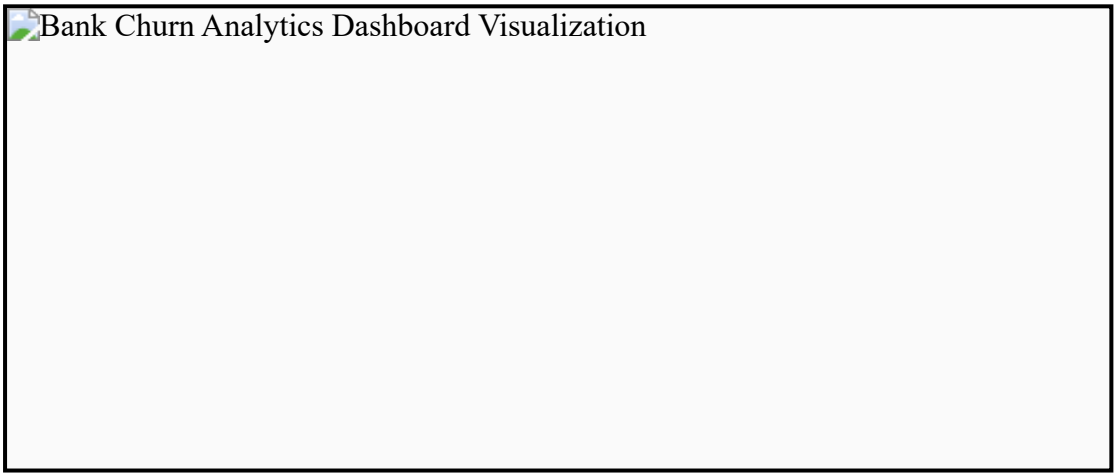
5. KEY INSIGHTS & OBSERVATIONS



1. The German Anomaly: While France and Spain remained stable (~16%), the German market experienced a massive capital exodus (32.44% churn).

2. Critical Zone (Age 46-60): Over half of the customers in their peak earning years exited, indicating a failure in wealth management or retirement products.

3. Premium Wealth Flight: High-balance customers churned at a significantly higher rate (24.16%) compared to zero-balance accounts (13.82%).



6. STRATEGIC RECOMMENDATIONS & CONCLUSION

Recommendations: Initiate an immediate operational audit of German branches. Deploy dedicated relationship managers for high-balance segments and overhaul financial products for the 46-60 demographic.

Conclusion: The project successfully translates raw banking data into targeted retention strategies, presented through a highly professional UI.

THANK YOU

I would like to express my sincere gratitude to **Unified Mentor** for providing me with this opportunity to learn and grow.

Working on this project has significantly enhanced my skills in **Data Analytics, Python, and UI/UX Dashboarding**. This internship has been a valuable experience in bridging the gap between theoretical knowledge and real-world application.

Final Report Submitted by

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